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Robustness of the Ensemble Score Filter to the Type of Assimilated Observation Networks

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ABSTRACT

Recent advances in data assimilation (DA) have focused on developing more flexible approaches that can better accommodate nonlinearities in models and observations. However, it remains unclear how the performance of these advanced methods depends on the observation network characteristics. In this study, we present initial experiments with the surface quasi-geostrophic model, in which we compare a recently developed ensemble filter using score-based diffusion models with the standard Local Ensemble Transform Kalman Filter (LETKF). Our results show that the analysis solutions respond differently to the number, spatial distribution, and nonlinear fraction of assimilated observations. We also find notable changes in the multiscale characteristics of the analysis errors. Given that standard DA techniques will eventually be replaced by more advanced methods, we hope this study sets the ground for future efforts to reassess the value of Earth observing systems in the context of newly emerging algorithms.

1 | Introduction

Data assimilation (DA) methods have made steady progress in the geosciences ever since the early days of objective analysis schemes (Daley 1991; Evensen et al. 2021; Kalnay et al. 2024). Advanced forms of ensemble and variational methods, along with hybrid approaches, now represent the state of the art in operational weather prediction centers (Bannister 2017). These algorithms efficiently assimilate large volumes of observations to define the initial conditions of complex models and enhance their forecast skill.

In parallel with these algorithmic advances, the Earth observing system has evolved significantly. For example, satellite observations are now routinely incorporated into models and have become a major contributor to improved analysis and forecast quality. In recent years, significant progress has been made in the assimilation of all-sky radiances (Geer et al. 2018)

and even data from alternative wavelengths, such as visible imagers (Scheck et al. 2020). At regional scales, assimilating high-frequency radar reflectivity and radial velocity data has demonstrated clear benefits for predicting convective storms (Dowell et al. 2011; Johnson et al. 2015; Gustafsson et al. 2018; Huang et al. 2022). Even in the historically under-sampled planetary boundary layer (Geerts et al. 2017), novel profiling data from ground-based remote sensors (Wulfmeyer et al. 2018; Cimini et al. 2020) and uncrewed autonomous systems (UAS; Barbieri et al. 2019; Bell et al. 2020) present new opportunities to improve the short-term forecasts of high-impact weather events (Chipilski et al. 2020, 2022; Degelia et al. 2020; Leuenberger et al. 2020; Jensen et al. 2021, 2022).

The value of the global observation network is continuously monitored at operational centers running numerical weather prediction models. Prior to the deployment of new instruments, comprehensive testing is often performed to evaluate their added

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value, either through data denial experiments (Kelly et al. 2007; Baker et al. 2022), or the generation of synthetic measurements using an Observing System Simulation Experiment (OSSE) framework (Arnold and Dey 1986; Masutani et al. 2007, 2010). These efforts have greatly improved our understanding of how various observation system components enhance the numerical forecast skill.

Nevertheless, one important limitation is that this understanding is largely tied to the standard ensemble and variational DA algorithms mentioned earlier. While these approaches are stable and computationally efficient for large geophysical applications, they rely on simplifying statistical assumptions (e.g., Gaussian error distributions) and often require linearization of the model and observation operators. This limits their accuracy when applied to high-resolution models (Poterjoy 2022) or complex nonlinear observations, such as all-sky radiances (Geer and Bauer 2011; Minamide and Zhang 2017). By contrast, there has been growing interest in advanced ensemble DA methods that offer more effective ways to blend sophisticated models and observations. These include nonlinear/non-Gaussian extensions of traditional methods (Amezcuca and van Leeuwen 2014; Tödter et al. 2016; Chan et al. 2020; Anderson 2022, 2023; Nerger 2022; Fletcher et al. 2023; Chipilski 2025), particle filter variants (Poterjoy 2016; Poterjoy et al. 2017; Potthast et al. 2019; van Leeuwen et al. 2019; Rojahn et al. 2023; Hu et al. 2024), and data-driven (AI-based) approaches (Arcucci et al. 2021; Boudier et al. 2023; Bocquet et al. 2024; Wang and Shen 2024; Hammoud et al. 2024; Bach et al. 2025; Manshausen et al. 2025). The previously mentioned studies have already demonstrated the potential benefits of these advanced DA methods across a range of spatiotemporal scales. However, their sensitivity to the underlying observation network design has not been adequately explored.

The purpose of this study is to emphasize how the choice of an ensemble DA method can generate distinct analysis responses to the type of assimilated observations, especially in terms of their multiscale characteristics. These impacts are examined using an intermediate-complexity model together with a newly developed ensemble data assimilation method based on score-based diffusion models (Bao et al. 2025). The present work can be viewed as a direct extension of Bao et al. (2025) given the choice of DA algorithms and numerical model. However, to our knowledge, this is the first systematic comparison of how different observation networks affect both traditional and newly developed DA methods. In particular, we demonstrate that diffusion-based ensemble DA methods are more robust to the assimilated network type compared to classical EnKF approaches.

Another finding with important practical consequences is that in highly turbulent dynamical systems, even a small fraction of nonlinear observations can make the optimal tuning of EnKF methods extremely challenging. The role of such nonlinear observations is expected to grow in the future, particularly in convective-scale applications, where high-frequency remote sensing systems (e.g., Doppler radars and ground-based profilers) play a crucial role in the analysis accuracy (Hu et al. 2023). This trend underscores the need for DA methods which remain robust under strongly nonlinear observation operators.

It should be noted that the presented numerical experiments follow some aspects of Hamill et al. (2002), who also used a quasi-geostrophic model and computed the power spectrum of the background-error variances as a function of horizontal wavenumber. However, our work extends this framework in two important ways. As mentioned earlier, we compare two different ensemble DA methods, rather than evaluating a single approach (3D-Var in their case). Second, we enhance the scope of the conducted experiments: in addition to varying the number of assimilated observations, we also examine the effects of their (i) spatial distribution and (ii) type (linear vs. nonlinear).

Ultimately, we hope that insights from our experiments will inspire more studies that incorporate a broader set of traditional and new DA methods, as well as more realistic geophysical models.

2 | Methods

To examine the sensitivity of different ensemble DA methods to the type of assimilated networks, we perform identical twin experiments which fit into the OSSE framework mentioned earlier. In this framework, synthetic observations are generated using the same numerical model that is later employed in the DA experiments (Kao et al. 2004).

2.1 | Numerical Model

For this study, we use the surface quasi-geostrophic (SQG) model, which exhibits turbulent characteristics similar to those of real geophysical flows (Rotunno and Snyder 2008). The SQG system assumes uniform potential vorticity, which is bounded between two horizontal surfaces separated by 10 km. We adopt the formulation of Tulloch and Smith (2009) which captures the nonlinear Eady dynamics. In this setup, the governing equations reduce to the advection of potential temperature on the two bounding surfaces.

These equations are integrated in time using a fourth-order Runge–Kutta scheme with a 20-min time step. Spatial derivatives are numerically approximated using a spectral collocation method based on the fast Fourier transform, taking advantage of the doubly periodic boundary conditions. All experiments are performed on a 64×64 grid on each of the two model surfaces, resulting in a model state dimension of 8192.

2.2 | Observing Networks

To perform identical twin DA experiments, we first initialize the SQG model with standard Gaussian noise and impose a sinusoidal pattern on the upper model surface. The SQG model is integrated for 300 days, with the first 100 days discarded to allow for sufficient model spin-up. The nature run (i.e., the true simulation) is defined as the model trajectory between days 100 and 150. During this period, synthetic observations are generated every 3 h, resulting in a total of 400 observation times. The observation generation process at a given time can be described using the stochastic model

$$\mathbf{y} = \mathbf{h}(\mathbf{x}_t) + \boldsymbol{\eta}, \quad \text{with } \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}). \quad (1)$$

where $\mathbf{x}_t$ is the true SQG state, $\mathbf{h}$ is the observation operator, and $\boldsymbol{\eta}$ is a Gaussian observation error realization with a zero mean and covariance $\mathbf{R}$. We consider two observation types:

- **Linear:** The observation operator $\mathbf{h}$ is a selection matrix that extracts values at a predefined set of observed grid points. The true SQG values at these locations are perturbed by additive Gaussian noise with covariance $\mathbf{R} = \mathbf{I}$. Both the observations and their associated errors are expressed in degrees Kelvin, as they represent potential temperature anomalies (see Figure 1 for a typical value range).
- **Nonlinear:** The observation operator $\mathbf{h}$ selects a subset of grid points and applies (component-wise) a nonlinear arctangent transformation: $\mathbf{h}(\mathbf{x}_t) = \arctan(\mathbf{x}_t)$. For this observation type, we use a covariance matrix with reduced observation error variances defined as $\mathbf{R} = 0.01^2 \times \mathbf{I}$. This scaling of the observation error noise compensates for the reduced range introduced by the arctan function. As shown in Figure 1, the true potential temperature values in the SQG model span from -25 to 25 K, whereas the arctan function is bounded within $(-\pi/2, \pi/2)$; e.g., see Figure 7 in

(Bao et al. 2025). The applied scaling ensures that the error standard deviations of the nonlinear observations remain consistent with the transformed SQG potential temperature values. Specifically, the error standard deviation is approximately 1% of the nature run's dynamical range, which is comparable to the 4% used in the linear observation case. Importantly, both ensemble DA algorithms tested in this study assimilate the same set of observations with identical error characteristics, enabling a consistent comparison of their responses to different observing networks.

In addition to varying the percentage of nonlinear observations (see Table 1), we also consider 3 observation networks with different spatial distributions, closely following the study of Morss et al. (2001). In the first two, observation locations are fixed in time: FIXED_EVEN uses evenly spaced observations, while FIXED randomly selects the observed grid points at the start of each DA experiment, keeping the same locations during the entire simulation. Figure 1 shows example distributions with 1024 observations from these two networks, overlaid on an example realization of the SQG state. The third network, referred to as RANDOM, has a spatial distribution similar to FIXED (Figure 1a) but changes the observation locations every analysis time. Thus, the RANDOM network can be viewed as a naive

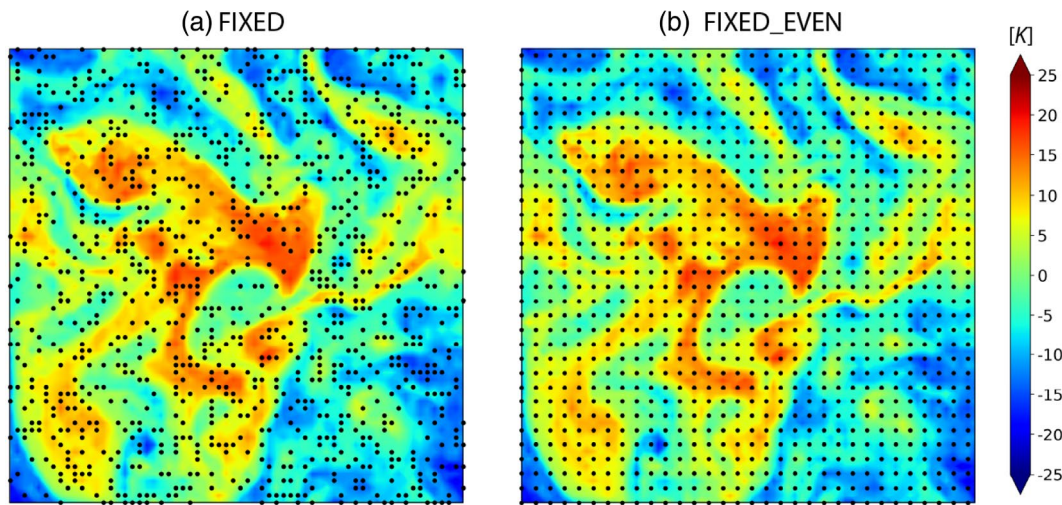


FIGURE 1 | Observation locations in the FIXED and FIXED_EVEN networks (black dots), overlaid on an example realization of the potential temperature anomaly from the nature run (color shading).

TABLE 1 | Time-averaged analysis RMSEs from LETKF and EnSF experiments assimilating 1024 observations (25% of all state variables).

Arctan%	LETKF			EnSF		
	FIXED	FIXED_EVEN	RANDOM	FIXED	FIXED_EVEN	RANDOM
0%	0.7	0.6	0.6	2.8	2.5	2.4
20%	10.1	8.0	9.0	2.8	2.5	2.4
40%	9.9	9.0	9.1	2.8	2.5	2.4
60%	9.6	9.5	9.7	2.8	2.5	2.5
80%	9.4	9.1	10.1	2.8	2.5	2.5
100%	9.2	9.0	9.2	2.8	2.6	2.5

Note: Each row represents a different percentage of nonlinear (arctangent) observations in the system.

or baseline adaptive sampling strategy, which has not been specifically optimized to reduce the forecast error growth (Morss et al. 2001).

2.3 | Ensemble Data Assimilation Methods

The initial ensemble for our DA experiments is formed by randomly selecting 20 states from the last 200 days of the true SQG integration (which spans a total of 300 days; see earlier discussion). The ensemble filters are evaluated over the same 50-day period during which the synthetic observations were generated, resulting in 400 analysis cycles. The primary objective of the numerical experiments presented in the next section is to evaluate the impact of different observation networks on both standard and recently developed ensemble filters.

For our reference DA approach, we employ the Local Ensemble Transform Kalman Filter (LETKF; Hunt et al. 2007), which is described in Appendix S1. Although LETKF assumes Gaussianity during the analysis step, its computational efficiency and localized formulation have made it one of the most widely used EnKF variants, including operational implementations at several numerical weather prediction centers (Schraff et al. 2016; Frolov et al. 2024).

To evaluate how more advanced nonlinear ensemble filters respond to changes in the assimilated observation networks, we consider the Ensemble Score Filter (EnSF; Bao et al. 2024, 2025). EnSF employs a Monte Carlo approximation of the posterior score, estimated from the forecast ensemble, and a user-specified likelihood model. The estimated posterior score is then used to guide a reverse diffusion process that transforms standard Gaussian samples into the desired analysis (posterior) ensemble. For a detailed description of the algorithmic workflow, readers are referred to Bao et al. (2025).

Although several other nonlinear filtering approaches were reviewed earlier, we focus on EnSF due to its demonstrated stability in the SQG setting (Bao et al. 2025). This robustness makes it a strong candidate for investigating the sensitivity to the different observation networks examined in this study.

3 | Results

Since the primary advantage of EnSF over LETKF lies in its ability to handle nonlinearities in the analysis step, we first examine the impact of increasing the fraction of nonlinear observations. Specifically, we consider a scenario in which 25% of the SQG model state is observed (1024 observations in total) and progressively increase the proportion of arctangent observations in increments of 20% across all three observation networks.

In the fully linear case (0% arctangent observations), Table 1 shows that LETKF is nearly four times more accurate across all networks. This finding is consistent with Bao et al. (2025), who demonstrated that a well-tuned LETKF retains clear advantages in an idealized, fully linear setup. These advantages are expected since LETKF is an ensemble approximation of the Kalman filter, which is optimal for linear-Gaussian systems. Its

strong performance is further reinforced by the fact that one of the current challenges for EnSF is handling partial observation networks, which is a topic of ongoing research.

Across different networks, we observe that the largest analysis mean errors occur in the FIXED network, whereas FIXED_EVEN and RANDOM achieve comparable performance. This pattern holds for both LETKF and EnSF, underscoring the importance of evenly sampling geophysical systems characterized by well-developed turbulence.

For all hybrid (linear-nonlinear) observation networks in Table 1, EnSF consistently yields visibly lower analysis RMSEs. Notably, the relative impacts of different networks remain unchanged: FIXED continues to yield the worst performance, while RANDOM and FIXED_EVEN achieve similar results. However, in all nonlinear LETKF experiments, FIXED_EVEN emerges as the optimal network, consistently outperforming RANDOM. The latter ranks as the second-best network in most nonlinear experiments, except in the case with 80% arctangent observations, where FIXED proves more beneficial.

The previous results indicate a shift in the behavior of the two ensemble filters for nonlinearities between 0% and 20%. This raises the question of identifying the threshold fraction of nonlinear observations at which LETKF begins to degrade. To address this, Figure 2 presents a detailed evolution of the analysis mean errors as the fraction of arctangent observations increases from 0% to 20% in increments of 5%.

The first key observation from the analysis RMSE time series in Figure 2 is that EnSF maintains consistent performance regardless of the degree of nonlinearity. On the other hand, the LETKF experiments confirm that the FIXED_EVEN network remains optimal in the presence of nonlinearities. With 20% arctangent observations, the RANDOM network slightly outperforms FIXED, as shown in Table 1. However, a new insight from the time series is that the RANDOM network results in a more rapid deterioration of LETKF's analysis accuracy. While LETKF retains its advantage over EnSF up to 15% observational nonlinearity in the FIXED and FIXED_EVEN networks, its performance in the RANDOM network begins to degrade as early as 5% arctangent observations.

Another consequence of increasing the degree of nonlinearity in the observing system is the growing difficulty of determining optimal tuning parameters for the benchmark LETKF algorithm. Figure 3 shows that in a fully linear observation network, there exists a broad range of localization and inflation parameters for which LETKF performs well (indicated by the blue regions). Evidently, multiple combinations of localization and inflation values yield comparable analysis errors. Furthermore, the tuning plots appear nearly identical across the three observation networks, suggesting that LETKF's tuning is largely insensitive to changes in the spatial distribution of observations. This is further supported by the fact that the optimal localization and inflation values remain similar across all three networks.

The second row of Figure 3 demonstrates that introducing even a small fraction (5%) of nonlinear observations significantly reduces the range of localization and inflation parameters for



FIGURE 2 | Evolution of the analysis RMSEs for LETKF (dashed lines) and EnSF (solid lines) with the FIXED, FIXED_EVEN, and RANDOM networks, each assimilating 1024 observations (25% of all state variables). Different colors indicate varying percentages of nonlinear (arctangent) observations.

which LETKF remains stable and avoids filter divergence. As an extreme case, in the RANDOM network (Figure 3c,f,i) with 5% nonlinearity, only a single parameter combination (a localization scale of 1000km and RTPS of 0.9) results in a well-behaved LETKF algorithm with sufficiently small RMSEs. As the fraction of nonlinear observations increases to 15%, tuning becomes even more difficult. For the RANDOM network, no combination of localization and inflation values prevents LETKF from experiencing large analysis RMSEs and filter divergence cases.

A completely different behavior emerges in the EnSF experiments. As shown by the solid curves in Figure 2, the RMSE time series exhibit minimal variability, both with respect to the fraction of nonlinear observations and the overall spatial distribution of the assimilated network. This robustness persists despite using a simple version of EnSF that is neither localized nor tuned for optimal inflation values (see Appendix S2).

Since the SQG model encompasses multiple scales of motion, it is worth analyzing how different observation networks influence the spectral characteristics of the analysis mean errors. The left panel of Figure 4 focuses on the scenario in which the entire observation network consists of linear observations.

Examining the LETKF results first, we confirm the expected trend that increasing the number of observations from 256 (6.25 % of all state variables) to 1024 (25 % of all state variables) induces lower analysis errors across all scales. However, the error reduction is most pronounced at the smallest wavenumbers (largest wavelengths), resulting in a modification of the overall spectral slope. In particular, the sparser observation network leads to more substantial analysis errors at large scales. According to the numerical experiments of Durran and Gingrich (2014), this scenario is more conducive to a rapid growth of the subsequent forecast errors. The effect arises from the forward cascade of energy in a turbulent regime characterized by a $-5/3$ power law, which applies to the SQG dynamics simulated in this study. Conversely, the denser networks with 1024 observations correspond to the opposite case: most of the analysis errors are concentrated within the small scales, which would imply a slower forecast error saturation.

Similar effects are observed in the EnSF experiments, although the overall sensitivity to the number of observations is smaller than that of the LETKF. For example, the spectral density happens to be nearly identical at the smallest wavelengths across all networks.

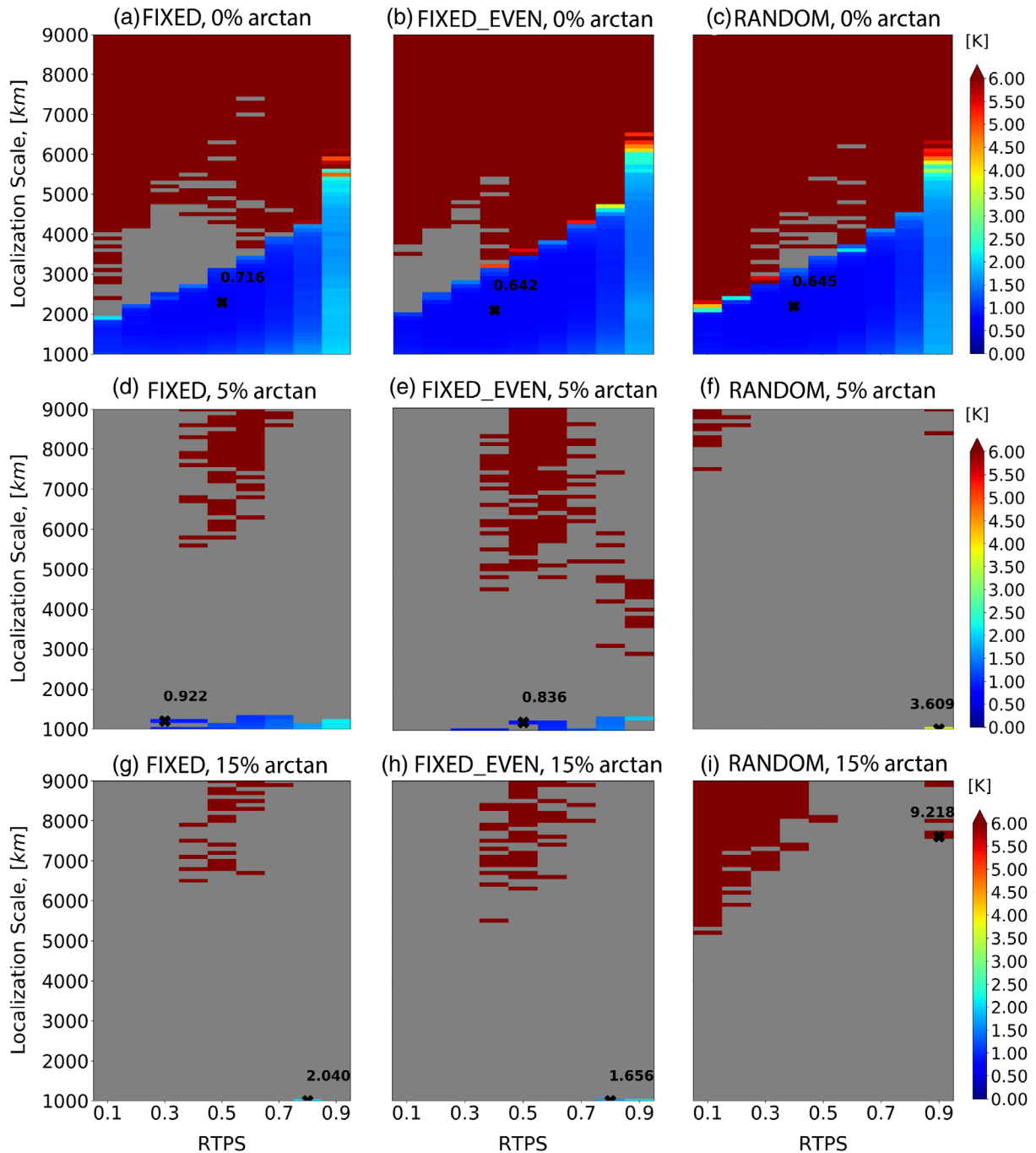


FIGURE 3 | Illustrating the challenges in optimally tuning the LETKF algorithm as the number of nonlinear (arctangent) observations increases. Each plot shows the time-averaged analysis RMSEs as a function of the RTPS (relaxation to prior spread) inflation parameter and horizontal localization scale. Rows correspond to different observation distributions in space (FIXED, FIXED_EVEN, and RANDOM), while columns indicate varying percentages of nonlinear (arctangent) observations. Similar to Figure 2, the total number of assimilated observations is 1024 (25% of all state variables).

The impacts caused by the spatial distribution of observations also reveal several insightful findings. For example, differences among FIXED, FIXED_EVEN, and RANDOM in LETKF (Figure 4) become more pronounced in the sparse observation regime. By contrast, the EnSF analysis errors exhibit a much clearer separation in the denser network case, particularly at larger scales. Nevertheless, a consistent finding across both filters is that the FIXED network yields the largest analysis errors regardless of how many observations are assimilated.

The spectral decomposition of the LETKF and EnSF analysis errors with 20% nonlinear (arctangent) observations is shown in panels (c) and (d) of Figure 4. Results are presented only for networks with 1024 observations (25% of all state variables) due to the divergent behavior of LETKF in sparser regimes. Consistent with previous findings, we observe minimal differences in the EnSF analysis results when nonlinear observations are assimilated. On the other hand, the nonlinear LETKF experiments exhibit stark differences in the magnitude of analysis errors as

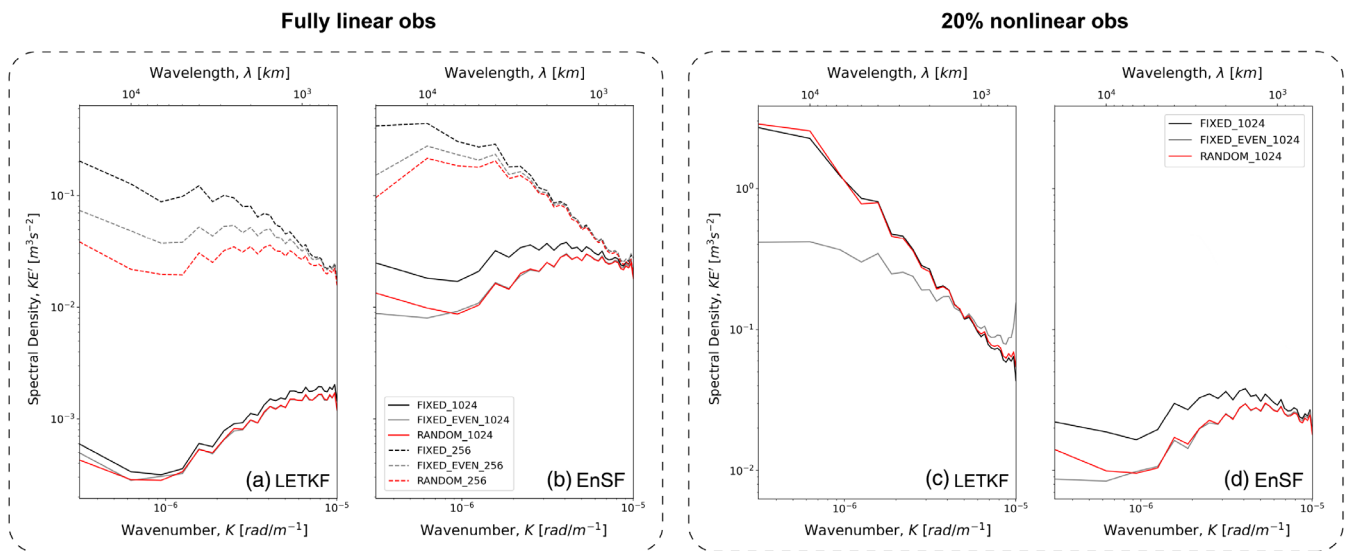


FIGURE 4 | Comparison of the multiscale impacts on LETKF and EnSF analyses through the kinetic energy spectrum of the analysis mean errors for a fully linear observation network (left dashed box) and a network containing 20% nonlinear (arctangent) observations (right dashed box). Solid (dashed) lines correspond to experiments assimilating 1024 (256) observations, respectively. The color of each spectral density curve indicates the spatial distribution of the assimilated network.

a function of scale. The resulting spectral density curves resemble those in the sparser observation regime shown in Figure 4a, with the most substantial analysis errors occurring at the largest scales of motion. Thus, in addition to the overall degradation of the LETKF performance, the slope of the kinetic energy spectrum would favor a more rapid error growth in subsequent SQG predictions (Durrán and Gingrich 2014).

4 | Conclusion

This study explored the sensitivity of two ensemble filters to variations in the assimilated observation networks: the standard Local Ensemble Transform Kalman Filter (LETKF) and the recently developed Ensemble Score Filter (EnSF), which leverages diffusion models to sample from complex analysis distributions. Numerical experiments were conducted using the surface quasi-geostrophic (SQG) system, which exhibits turbulent characteristics similar to those of real geophysical flows.

We compared both ensemble filters with respect to several key aspects of the assimilated observation networks, including density, spatial distribution, and the fraction of nonlinear observations in the system. A common finding across the two DA methods was that a fixed network with evenly distributed observations (FIXED_EVEN) resulted in smaller analysis errors compared to an inhomogeneous network (FIXED), which is more representative of real Earth observing systems. One of the novel aspects of this finding lies in its generalization across the chosen DA algorithm (both LETKF and EnSF) as well as the degree of nonlinearity in the observation system. By contrast, a network of randomly distributed observations (RANDOM) exhibited different impacts depending on the chosen DA method. For the diffusion-based EnSF, it yielded analysis skill comparable to the well-performing FIXED_EVEN network, while the RANDOM network in LETKF often performed worse relative to the other two configurations.

Although it is well known that increasing the number of observations improves the analysis quality, our experiments provided new insight into how observation density alters the spectral characteristics of the analysis errors. Notably, we observed changes in the slope of the kinetic energy spectrum, which has implications for the growth of initial condition uncertainty (Durrán and Gingrich 2014; Rotunno and Snyder 2008).

Gradually increasing the fraction of nonlinear observations also led to important consequences. While an optimally tuned LETKF performed well for a fully linear observation network, the benefits of the more flexible EnSF became apparent even when only a small fraction of the assimilated observations were nonlinear (5% in the RANDOM network case). The presence of nonlinear observations was also shown to alter the spectral characteristics of the LETKF analysis errors, leading to visible increases at the large scales. On the contrary, EnSF maintained consistent performance across all network configurations, despite the absence of explicit localization or inflation tuning.

There are several practical implications of our results for operational NWP systems, as many of them rely on variants of the EnKF. For example, the German Kilometer-scale ENsemble-based Data Assimilation (KENDA) system (Schraff et al. 2016) is based on the LETKF approach to provide convective-scale DA analyses. Our LETKF tuning experiments (Figure 3) suggest that it becomes increasingly more difficult to identify optimal localization and inflation parameters when nonlinear observations are assimilated. According to personal correspondence with KENDA developers, radar reflectivity data account for just over 20% of the observations in their DA system, based on statistics from February 11, 2025. These observations are highly nonlinear and conceptually resemble the arctangent observation operator used in our study. Although their current DA system remains stable, the projected growth in other nonlinear observations (e.g., all-sky radiances) may challenge LETKF's effectiveness in assimilating such data. This highlights the need for additional case

studies evaluating the performance of LETKF and other EnKF variants in highly nonlinear observational regimes.

The reduced sensitivity of EnSF to both the spatial distribution of observation networks and the degree of nonlinearity (e.g., Figure 2) raises the broader question of whether such robustness is a general property of other nonlinear DA methods. A systematic exploration of this topic in realistic geophysical systems like KENDA would be of practical value, as it will offer guidance for the design of future observation networks. Moreover, the development of new impact metrics that more accurately account for nonlinear effects will be necessary to better evaluate and prioritize these emerging observation types.

Author Contributions

Zixiang Xiong: software, data curation, investigation, validation, formal analysis, visualization, writing – original draft, writing – review and editing. **Siming Liang:** software, data curation, supervision, writing – review and editing. **Feng Bao:** conceptualization, supervision, funding acquisition, writing – review and editing. **Guannan Zhang:** conceptualization, supervision, writing – review and editing. **Hristo G. Chipilski:** conceptualization, methodology, software, investigation, supervision, funding acquisition, writing – original draft, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in GitHub at <https://github.com/ZixiangXiong/sqgpublic>.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** Supporting Information.